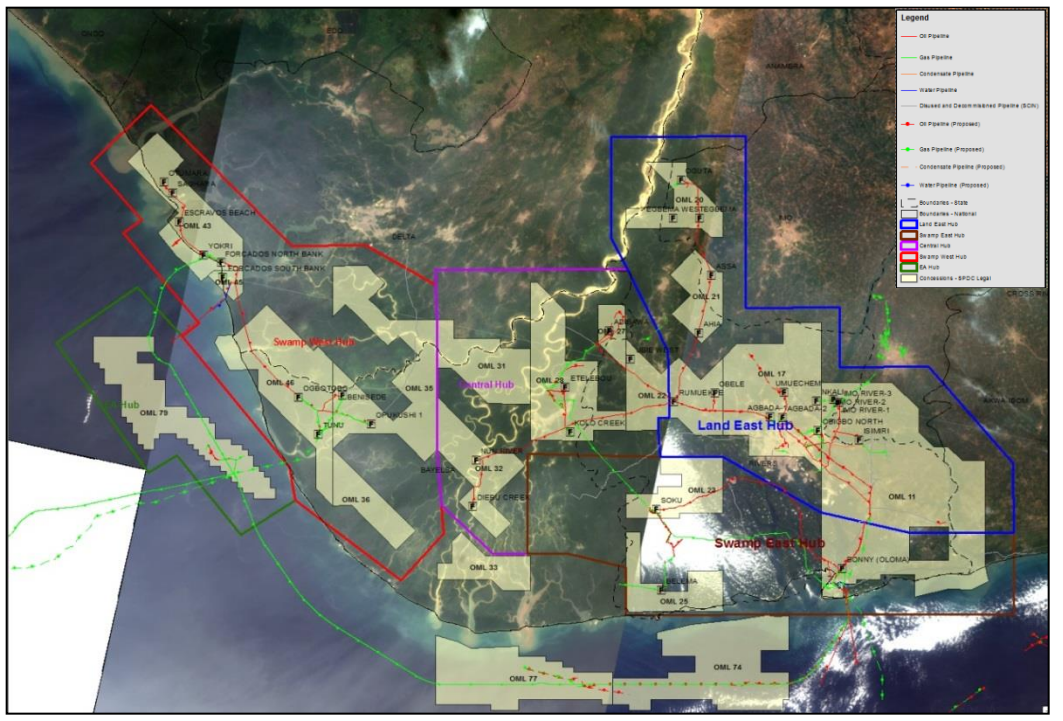
SCiN-WIDE DESKTOP PIPELINE RIGHT OF WAY ENCROACHMENT STUDY

SCIN DELTA-WIDE FOOTPRINT



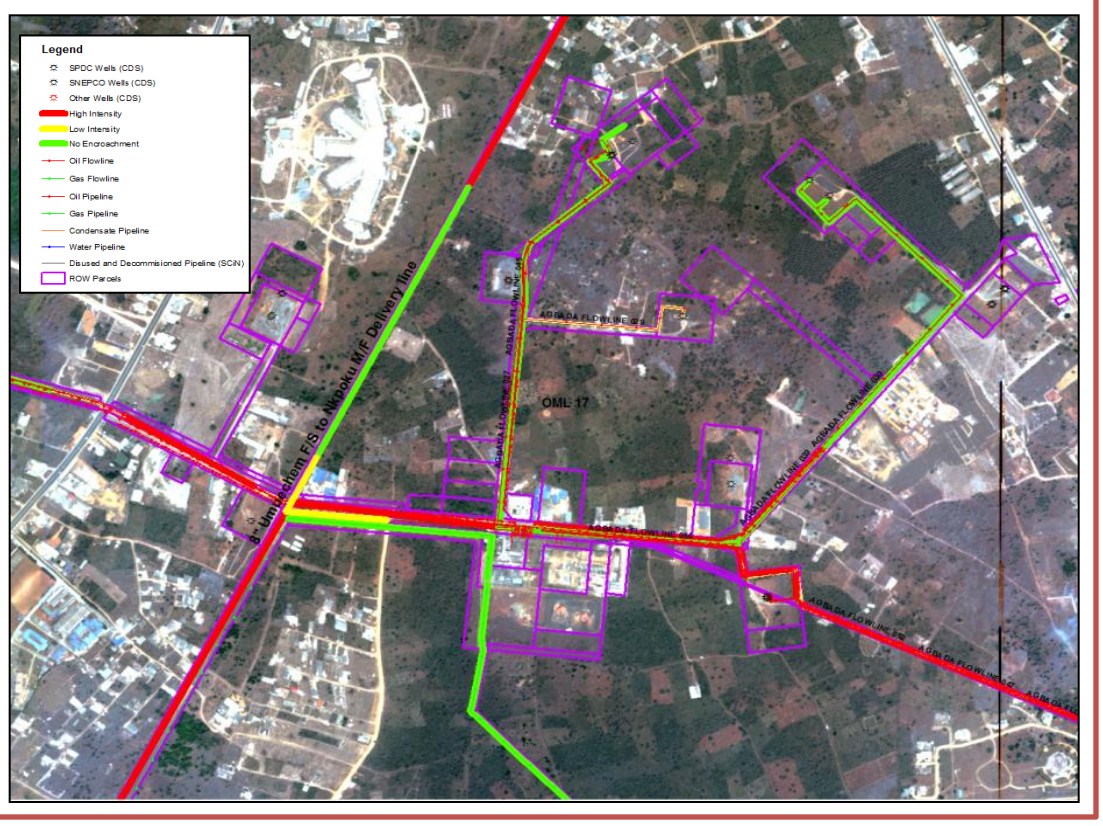
CLIP SHOWING SPATIAL ANALYSIS CONCEPT

Table

Encroachment Points

OBJECTID *	Shape *	Certainty	Point_Number	Latitude	Longitude	Ref_Facility	Encroachment_Description	Cocncession	Attributes_Updated_By	Attributes_Updated_Date	Geom_Updated_By	Geom_Updated_Date
20496	Point	Possibly	PT20496	436424.3	112604	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20497	Point	Possibly	PT20497	436435.6	112825.3	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20503	Point	Possibly	PT20503	436525.5	112763.6	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20506	Point	Possibly	PT20506	436560.3	112821	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20509	Point	Possibly	PT20509	436600.8	112880.5	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20527	Point	Possibly	PT20527	436682	113252.2	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20530	Point	Possibly	PT20530	436759.9	113069	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20535	Point	Possibly	PT20535	435153.9	109514.6	Pipelines	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20539	Point	Possibly	PT20539	434925.8	109551	Pipelines	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20561	Point	Possibly	PT20561	425933.3	111301.6	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20565	Point	Possibly	PT20565	425975.2	111462.2	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20566	Point	Possibly	PT20566	425942.3	111279	Others	Within 10m-15m RoW Boundary Buffer	OML28	Shebolatan Babatunde	04/07/2018	Shebolatan Babatunde	04/07/2018
20580	Point	Possibly	PT20580	516905.3	99136.7	Others	Within 10m-15m RoW Boundary Buffer	OML17	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
20581	Point	Possibly	PT20581	516853.8	99179.48	Others	Within 10m-15m RoW Boundary Buffer	OML17	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
20582	Point	Possibly	PT20582	516894.8	99150.7	Others	Within 10m-15m RoW Boundary Buffer	OML17	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
20583	Point	Possibly	PT20583	516882.6	99139.66	Others	Within 10m-15m RoW Boundary Buffer	OML17	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
20600	Point	Possibly	PT20600	523310.1	94971.16	Others	Within 10m-15m RoW Boundary Buffer	OML11	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
20601	Point	Possibly	PT20601	523322.5	94973.46	Others	Within 10m-15m RoW Boundary Buffer	OML11	Oyetunde Simisola	04/07/2018	Oyetunde Simisola	04/07/2018
1	Point	Probably	PT1	467761	87915.73	Others	Within 10m RoW Boundary Buffer	OML23	Ojeh Amaechi	03/07/2018	Ojeh Amaechi	03/07/2018
2	Point	Probably	PT2	467750.9	87887.87	Others	Within 10m RoW Boundary Buffer	OML23	Ojeh Amaechi	03/07/2018	Ojeh Amaechi	03/07/2018
3	Point	Probably	PT3	467740.3	87856.05	Others	Within 10m RoW Boundary Buffer	OML23	Ojeh Amaechi	03/07/2018	Ojeh Amaechi	03/07/2018
4	Point	Probably	PT4	467763.5	87862.57	Others	Within 10m RoW Boundary Buffer	OML23	Ojeh Amaechi	03/07/2018	Ojeh Amaechi	03/07/2018

CLIP SHOWING ENCROACHMENT INTENSITY CONCEPT



INTRODUCTION

SCiN has its footprint of pipeline network and oil and gas facilities spanning the Niger-Delta within host communities. Over the years there has been gradual intrusion (Encroachment) of settlements into the acquired Right Of Way (ROW) for our existing facilities (pipelines, flowlines, flow stations, wells etc.). The proximity of these settlement structures (e.g. buildings) to our facilities pose a great risk in terms of health ,Safety and Security (HSE) and also is a great threat to the continuous production from our facilities as it is a major breach in our planned safety standards.

It is imperative that we are aware of the level of encroachment along our ROW so as to carry out necessary actions to reduce this exposure. In view of large number of pipeline decommissioning activities planned in the near future, and associated significant liability of circa **$7.48bln** for Facilities, Pipeline and Borrow Pit there is a current drive to estimate SPDC Asset Retirement Obligation inclusive of the cost of decommissioning the entire pipelines within SPDC (Both non-operational & producing). Based on regulatory steers, the cost for decommissioning our existing pipelines will differ based on the level/intensity of encroachment into the respective ROW thus the need to estimate the degree of encroachment using Remote Sensing and GIS techniques.

OBJECTIVES

- To determine the degree/intensity and form of encroachment along all existing SCiN pipelines delta -wide with the aim of possibly avoiding/reducing HSE exposure in mobilizing personnel for full scale survey in view of the current security sensitivities.
- To determine the extent beyond a pipeline ROW boundary that is required to be free of 3rd party habitation for safe de-oiling & removal of a pipeline within the ROW.

WORK SCOPE

- Satellite Image Processing
- Desktop analysis and encroachment Classification
- Geomatics Spatial database creation
- Publication of GIS layers
- Paper Based and WebMap Creation
- Report Delivery
- Project Close-out session with stakeholders.

METHODOLOGY

- Analysis of high resolution optical and radar remote sensing satellite images in conjunction with spatially referenced Legal Right Of Way (ROW),Pipelines, Facility Locations (i.e. flow stations, Gas plants, Wells and Manifolds) and 15m ROW buffer encroachment zone.
- Mapping and classification of encroachment points based on certainty and intensity.
- Proper data Archiving/Storage
- Readily Available Time Lapse data.
- Result served via a MyMaps portal thereby providing a credible database for mitigation/repair/placement of ROW markers.

ASSUMPTIONS FOR STUDY

- **Encroachment Certainty (Degree of how sure you are that there is an encroachment):**
  - Probably on Pipeline i.e. encroachment point is < 10m of R0W boundary
  - Possibly on Pipeline i.e. encroachment point is >10m and <15m (Beyond 10m but within 15m) of ROW boundary
- **Encroachment Intensity (Encroachment along ROW corridor categorised based on number points per 10meters)**
  - High Intensity means that > 10 encroachment points per 1km was picked
  - Low Intensity means that < 10 encroachment points per 1km was picked
  - No Encroachment means zero encroachment points per 1km was picked

****Note: Due to inherent limitation in satellite positioning and remote sensing processing data accuracy ,ground truthing is imperative for the determination of the actual position of encroaching structures.**

BUSINESS VALUE

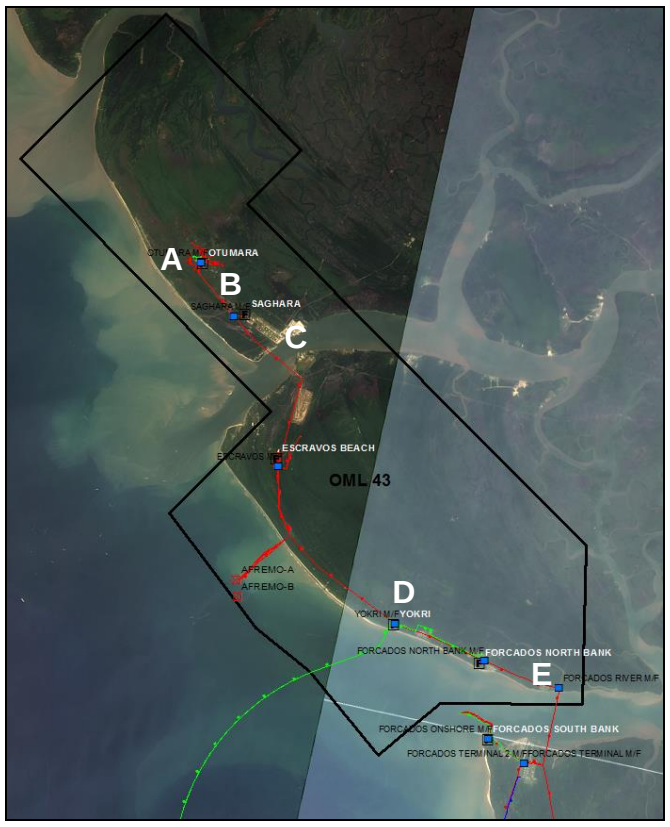
- **Cost saving** – This approach was able to demonstrate a reduction of **circa $50mln** of SPDC D&R liability out of the potential $1.1bln earmarked for reduction in 2018
- **Risk reduction** – significant reduction in exposure of personnel and equipment to hazardous and challenging land and swamp environments with clarity on extent of encroachment into SPDC Right Of Way and effective mitigations.
- **Situational Awareness** – Up-to-date, readily available situational awareness for timely engagement with government stakeholders, planning and informed decision making.

Report By: Oyetunde Simisola PTD/E/N

Project Team: Obowu Chituru PTD/E/N, Oyetunde Simisola PTD/E/N , Ojeh Amaechi PTD/E/N, Omekara Okwuchi PTD/E/N , Shebolatan Babatunde PTD/E/N

OML 43 DESKTOP ENCROACHMENT STUDIES

OML 43



INTRODUCTION

OML 43 concession which is ~ 768Sqkm in area covers considerable part of Warri North and Warri South West Local Government areas along the North-West Part of the Nigerian shoreline.

This concession hosts 6 manifolds (Otumara, Sghara, Yokri, Forcados North, Escravos Beach and Forcados River Manifolds) and 5 flow stations (Otumara, Saghara, Yokri, Forcados North and Escravos Beach Flow Stations) with a network of ~88km length of pipelines and flowlines connecting the facilities. The existing pipeline /flowline network crosses host communities in which buildings are clustered around the right of way for the pipelines and flowlines.

In view of the future plan to decommission these pipelines and flowlines it is important to deduce the length of pipeline ,area of right of way and host settlements affected by high or low intensity encroachment. It is also important that the results of the analysis feed into the estimation of the cost of decommissioning these pipelines to ensure proper project value estimation and planning in line with the tru ground situation along the right of way within the concession.

LIST OF PIPELINES AND FLOWLINES WITHIN ENCROACHED AREAS

- Rapele - Forcados T/L - 50.0/28“
- Escravos Forcados T/L - 26.4/24“
- Trans Escravos T/L (OTUMARA MFD - ESCRAVOS MFD) - 17.1/20“
- Otumara DL (OTUMARA F/S - OTUMARA MFD) -0.1/8“
- Saghara D/L (SAGHARA F/S - SAGHARA MFD) -2.0/8"
- Escravos Forcados T/L - 26.4/24“
- 24" Export Gasline
- 16" Associated Gasline
- Forcados Yokri Flowline 115T - /1.800KM
- Forcados Yokri Flowline 113T - /3.900KM
- Otumara Flowline-022
- Otumara Flowline-005l
- Otumara Flowline-005s
- Otumara Flowline-032t
- Saghara 002
- Saghara 001
- Saghara 007
- Saghara 008
- Saghara 003
- Saghara 006

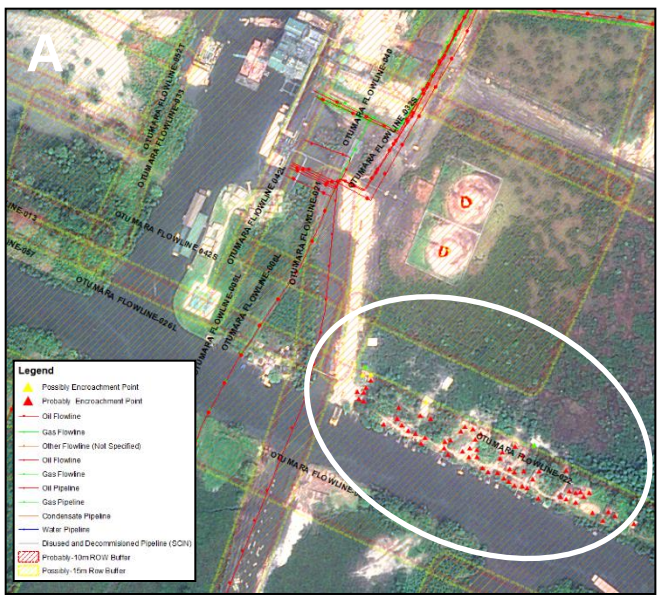


Figure 4: Clip Showing Encroachment Area A

Figure 5: Clip Showing Encroachment Area B

Figure 6: Clip Showing Encroachment Area C

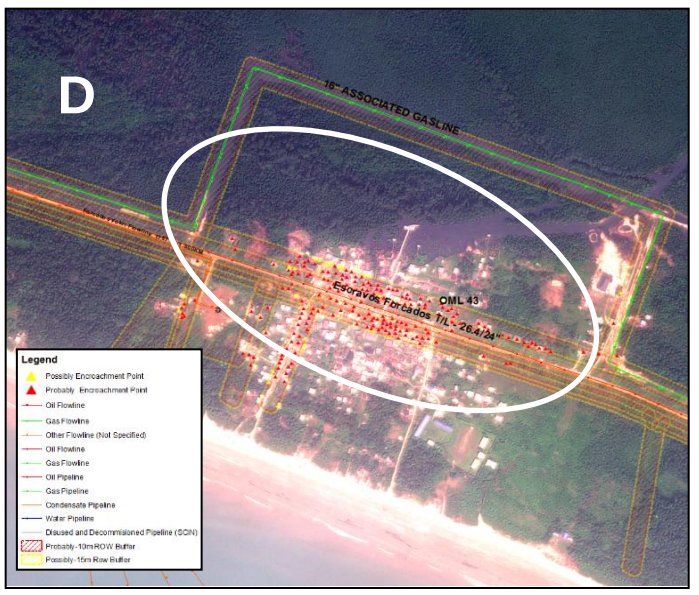


Figure 7: Clip Showing Encroachment Area D

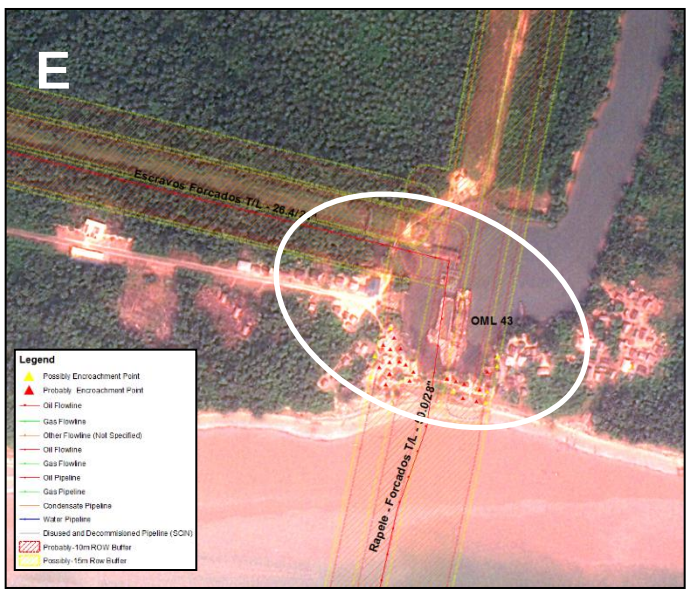


Figure 8: Clip Showing Encroachment Area E

STATISTICAL ANALYSIS OML 43	
Total Pipeline Right Of Way Length Per Concession	107.28Km
Number Of Probably Encroached Points Per Concession	317
Number Of Possibly Encroached Points Per Concession	45
Total Number Of Encroached Points Per Concession	362
Probably Pipeline Right Of Way Length Encroached Per Concession	7.63Km
Possibly Pipeline Right Of Way Length Encroached Per Concession	1.08Km
Total Pipeline Right Of Way Length Encroached Per Concession	8.71Km
% Pipeline Right Of Way Length Probably Encroached	7.11%
% Possibly Right Of Way Length Probably Encroached	1.01%
% Total Right Of Way Length Encroached Per Concession	8.12%
Total Pipeline Right Of Way Area Encroached Per Concession	1.52Sqkm
Pipeline ROW high Intensity Area Encroached Per Concession	0.44Sqkm
Pipeline ROW Low Intensity Area Encroached Per Concession	0.43Sqkm
Total Pipeline Right Of Way Area Encroached Per Concession	1.52Sqkm
% Pipeline Right Of Way Area Probably Encroached Per Concession	4.54%
% Pipeline Right Of Way Area Possibly Encroached Per Concession	4.53%
% Total Right Of Way Area Encroached Per Concession	9.07%

LIST OF SETTLEMENTS WITH POSSIBLE OR PROBABLY ENCROACHED PIPELINES/FLOWLINES

- Uba Oke
- Aruton
- Bombelenge
- Yokri Sobo
- Abara

****Note: Due to inherent limitation in satellite positioning and remote sensing processing data accuracy ,ground truthing is imperative for the determination of the actual position of encroaching structures**

REMOTE SENSING ANALYST'S COMMENT:

From the study and spatial analysis approximately 8.71km of SCiN Pipeline Right Of Way length and approximately 1.52Sqkm of SCiN Pipeline Right Of Way Area has probably or possibly been encroached upon (i.e. encroachment points are within 10m to 15m buffer away from legal Right of Way) and there is significant encroachment along Rapele - Forcados T/L - 50.0/28“, Otumara Flowline-022, Saghara 003, Trans Escravos T/L (OTUMARA MFD - ESCRAVOS MFD) - 17.1/20 and Escravos Forcados T/L - 26.4/24“ with less than (>) or greater than (<) 5 picked every 1km within the encroached area.

Report By: Oyetunde Simisola PTD/E/N

Project Team: Obowu Chituru PTD/E/N, Oyetunde Simisola PTD/E/N , Ojeh Amaechi PTD/E/N, Omekara Okwuchi PTD/E/N , Shebolatan Babatunde PTD/E/N

SCiN-WIDE DESKTOP PIPELINE RIGHT OF WAY ENCROACHMENT STUDY RESULT

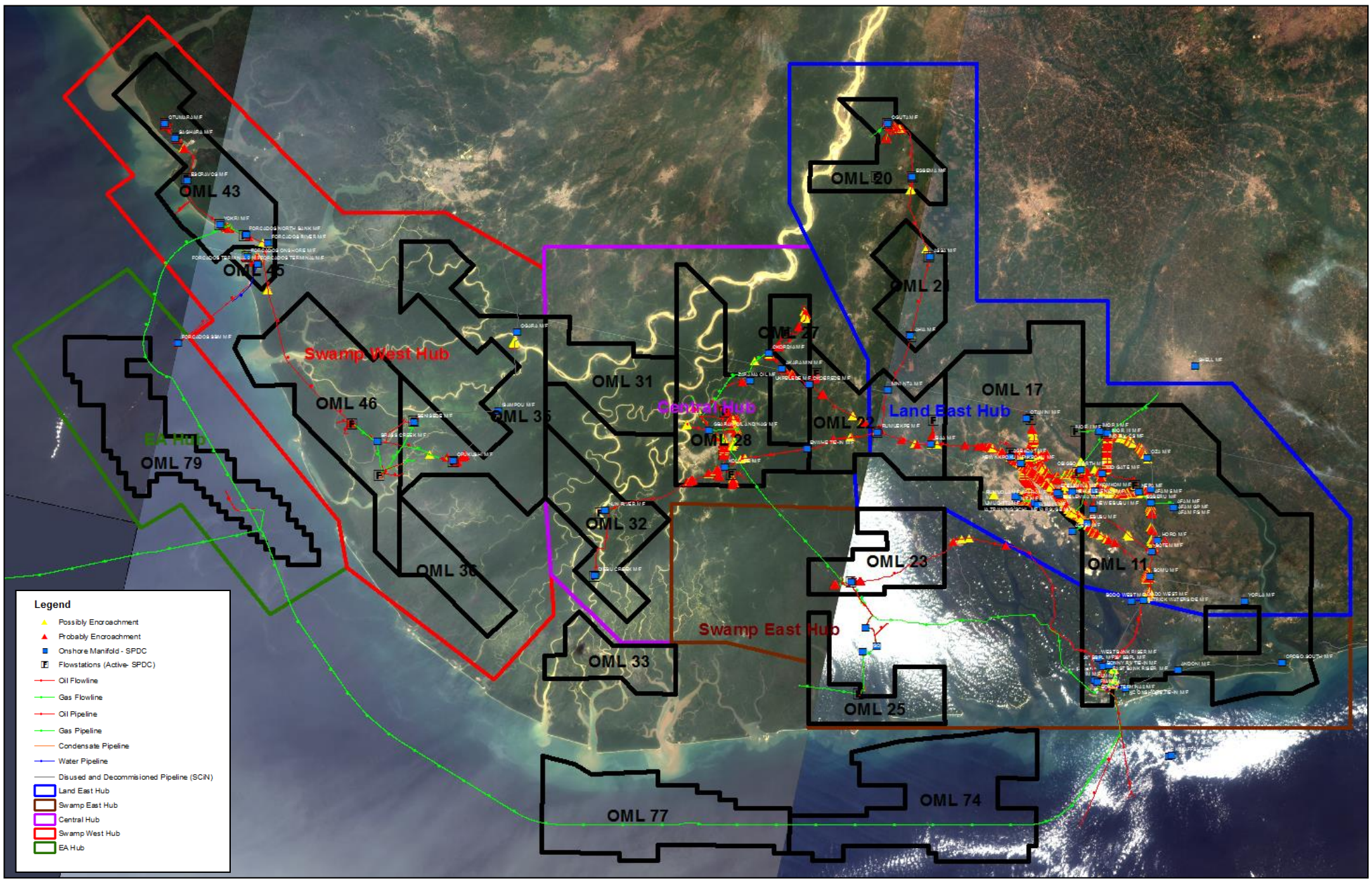
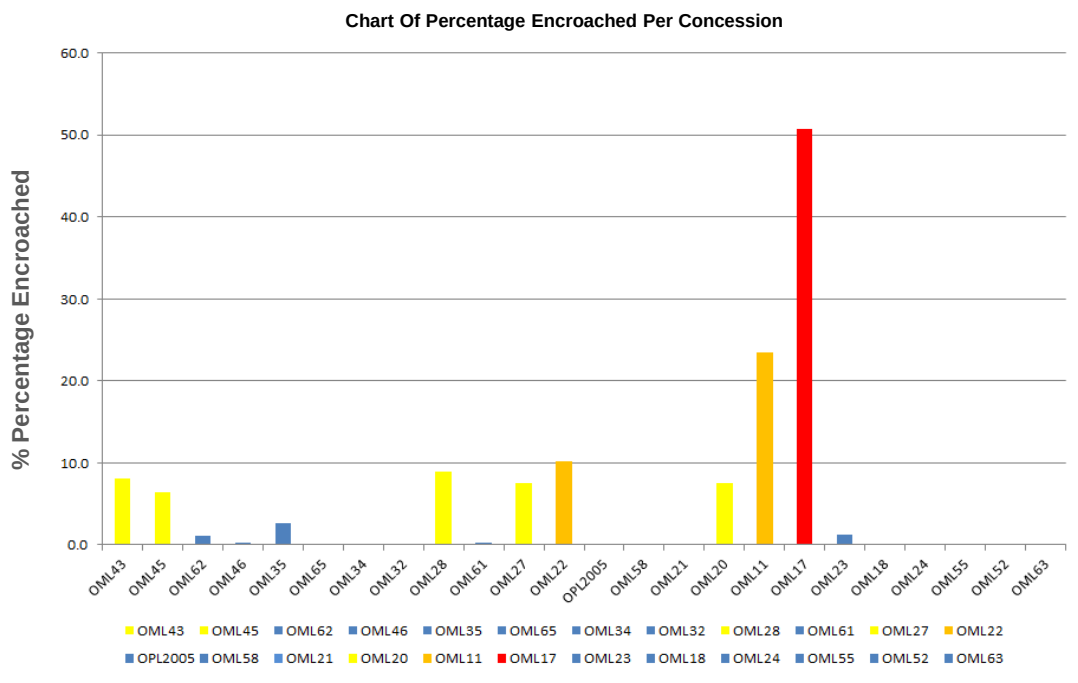


Figure 1: Clip Showing SCiN-Wide Desktop Encroachment Study Spatial Data Output

S/N	Concession	Number of Probably Encroached Points Per Concession	Number of Possibly Encroached Points Per Concession	Pipeline ROW high Intensity Area Encroached Per Concession (SqKm)	Pipeline ROW high Intensity Length Encroached Per Concession (Km)	Pipeline ROW Low Intensity Area Encroached Per Concession (SqKm)	Pipeline ROW Low Intensity Length Encroached Per Concession (Km)	Total Pipeline Right Of Way Area Encroached Per Concession (SqKm)	Total Pipeline Right Of Way Length Encroached Per Concession (Km)	%Pipeline Right Of Way Area Probably Encroached Per Concession	%Pipeline Right Of Way Length Probably Encroached	%Pipeline Right Of Way Area Possibly Encroached Per Concession	%Possibly Right Of Way Length Probably Encroached	% Right Of Way Area Encroached Per Concession	% Right Of Way Length Encroached Per Concession
1	OML43	317	45	0.44	7.63	0.43	1.08	1.52	8.71	4.54	7.11	4.53	1.01	9.07	8.12
2	OML45	603	21	0.64	3.19	0.04	0.11	0.15	3.31	8.23	6.22	0.48	0.22	8.72	6.44
3	OML62	3	11	0.00	0.13	0.00	0.48	0.48	0.61	0.00	0.25	0.40	0.92	0.40	1.17
4	OML46	16	0	0.08	0.36	0.00	0.00	0.00	0.36	1.04	0.33	0.00	0.00	1.04	0.33
5	OML35	12	12	0.03	1.42	0.02	1.42	1.45	2.85	0.40	1.35	0.32	1.35	0.72	2.70
6	OML65	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	OML34	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	OML32	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	OML28	512	76	1.03	17.55	0.23	2.61	2.83	20.15	10.61	7.79	2.33	1.16	12.94	8.95
10	OML61	40	19	0.04	0.24	0.01	0.11	0.12	0.36	7.35	0.20	1.89	0.09	9.23	0.29
11	OML27	92	`	0.06	2.70	0.02	0.50	0.52	3.20	4.96	6.40	1.73	1.18	6.68	7.58
12	OML22	70	24	0.09	5.89	0.49	2.02	2.51	7.91	4.80	7.60	25.97	2.60	30.76	10.20
13	OPL2005	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14	OML58	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15	OML21	3	2	0.00	0.04	0.00	0.02	0.03	0.06	0.00	0.11	0.27	0.07	0.27	0.18
16	OML20	3	2	0.38	2.34	0.02	1.56	1.58	3.91	20.66	4.56	0.84	3.04	21.49	7.59
17	OML11	120	24	1.83	99.28	0.43	19.86	20.29	119.14	9.51	19.53	2.23	3.91	11.74	23.43
18	OML17	1902	1066	2.05	58.23	1.62	32.64	34.26	90.87	23.81	32.56	18.86	18.25	42.68	50.81
19	OML23	3326	1106	0.29	0.62	0.00	0.21	0.21	0.83	4.70	0.93	0.00	0.31	4.70	1.24
20	OML18	1	134	0.00	0.00	0.00	0.12	0.12	0.12	0.00	0.00	0.13	0.07	0.13	0.07
21	OML24	4	3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
22	OML55	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23	OML52	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
24	OML63	0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Figure 2: Statistical Analysis Results Per Concession For SCiN-Wide Desktop Encroachment Study Spatial Data Output



REMOTE SENSING ANALYST'S COMMENT:

From the study and spatial analysis approximately 262km of SCiN Pipeline Right Of Way length and approximately 94Sqkm of SCiN Pipeline Right Of Way Area has probably or possibly been encroached upon (i.e. encroachment points are within 10m to 15m buffer away from legal Right of Way) and there is significant high/low intensity encroachment within OML 11,OML 17,OML28,OML22,OML45 and OMl23 with less than (>) or greater than (<) 5 picked every 1km within the encroached area.

	Value	Unit
Total Pipeline Right of way Length(Delta Wide)	2185.70	Km
Total Pipeline Right of way Length Encroached (Delta Wide)	262.38	Km
% Pipeline Right of way Length Encroached (Delta Wide)	12.00	%
Total Pipeline Right of Way Area (Delta Wide)	94.14	Sqkm
Total Pipeline Right of Way Area Encroached (Delta Wide)	10.28	Sqkm
% Pipeline Right of Way Area Encroached (Delta Wide)	10.92	%

****Note: Due to inherent limitation in satellite positioning and remote sensing processing data accuracy ,ground truthing is imperative for the determination of the actual position of encroaching structures.**

Report By: Oyetunde Simisola PTD/E/N

Project Team: Obowu Chituru PTD/E/N, Oyetunde Simisola PTD/E/N , Ojeh Amaechi PTD/E/N, Omekara Okwuchi PTD/E/N , Shebolatan Babatunde PTD/E/N